

Galactic Rotation Curves as Dark Matter Evidence

- **Freeman (1970, ApJ; observational)** – Early rotation-curve study of M33 and NGC 300. Freeman showed that the rotation speed peaks at about 2.2 disk scalelengths and *does not decline* as expected for the luminous disk alone ¹. In fact, no drop at 2.2 h was seen in these galaxies, implying a large extended “halo” of unseen mass. This paper is often cited as the first clear indication (beyond the luminous disk) that an extended dark component is needed ¹.
- **Bosma (1981, AJ; observational)** – 21-cm HI survey of ~25 spirals. Bosma found that in *every* case the HI rotation curve stays flat out to the last measured point. As Noordermeer *et al.* (2007) note, these flat HI curves (continuing well beyond the optical disk) provided “final, irrefutable evidence” for large amounts of unseen mass in galaxies ². (Rubin, Ford & Thonnard in the late 1970s found the same result optically for high-luminosity spirals, and Begeman *et al.* 1991 later confirmed it in dwarfs – all requiring massive dark halos.)
- **Persic, Salucci & Stel (1996, MNRAS; observational)** – Analysis of ~1100 spiral rotation curves. They developed the “Universal Rotation Curve” (URC) model, showing that rotation-curve shape correlates with galaxy luminosity. Bright galaxies are nearly disk-dominated (only a small dark component), whereas faint galaxies are strongly dark-matter dominated. They quantified scaling laws: e.g. the dark/visible mass ratio at the optical radius scales $\propto L^{-1}$, halo core radii ~optical radii, central halo density $\propto L^{-0.7}$, etc ³. In short, Persic *et al.* showed observationally that lower-luminosity disks require progressively more dark halo mass ³.
- **Noordermeer *et al.* (2007, MNRAS; observational)** – HI + optical rotation curves for 19 early-type (S0–Sab) disks. They found that many high-luminosity spirals have rotation curves that **decline slightly** at intermediate radii but *flatten out again* at large radii, implying substantial dark halos. As they report, “at large radii, all declining rotation curves flatten out, indicating that substantial amounts of dark matter must be present in these galaxies too” ⁴. They emphasize that the early HI work of Bosma (1981) and others in the 1970s – flat curves to the last point – was the “final, irrefutable evidence” for dark halos ².
- **de Blok *et al.* (2008, AJ; observational)** – High-resolution HI rotation curves from the THINGS survey (19 nearby disks). These curves have exquisite spatial and velocity resolution and extend far into the outskirts. de Blok *et al.* find *no* example of a definitive Keplerian falloff or “cutoff” out to the last measured point ⁵. They combine these HI curves with Spitzer infrared photometry to build mass models. Their key findings: massive, high-surface-brightness disks can be fitted either with cuspy (CDM-like) or cored halos equally well, but low-mass (dwarf) galaxies **strongly prefer cored halos**. Also, the derived dark halo densities are about half of what CDM simulations predict (independent of IMF choice) ⁵.
- **Lelli, McGaugh & Schombert (2017, ApJ; observational)** – The SPARC sample of 240 galaxies (spirals, irregulars and dwarfs) with accurate rotation curves and Spitzer photometry. They uncovered the tight *radial acceleration relation* (RAR): the observed centripetal acceleration a_{g}

$v_{\text{obs}} = V^2/r$ correlates extremely well with that predicted by the baryons g_{bar} (from the measured stellar+gas mass distribution). Galaxies of all types lie on one relation: at high g_{bar} the data follow the 1:1 line (no discrepancy), but below a critical scale $\sim 10^{-10} \text{ m/s}^2$ the relation systematically deviates, indicating extra acceleration. The scatter around this RAR is very small (<0.13 dex). In other words, given the baryonic mass profile, one can predict the rotation curve – but Newtonian dynamics fails below $\sim 10^{-10} \text{ m/s}^2$ ⁶. This empirical law encapsulates the flat-curve phenomenon and its dependence on baryons in a single relation ⁶.

- **Sofue & Rubin (2001, *ARA&A*; review)** – Authoritative review of spiral galaxy rotation curves. They emphasize that rotation curves are the primary tool for diagnosing galactic mass distributions ⁷. Their summary: essentially all well-measured spirals have rotation curves that stay flat (or nearly so) well beyond the optical disk, inconsistent with the light distribution. They discuss methods of measurement and the dependence of curve shape on Hubble type, surface brightness, etc., noting that none of these trends negate the basic need for a dark halo. (This review collects decades of data and confirms that flat rotation curves are a ubiquitous phenomenon.)
- **Milgrom (1983, *ApJ*; theoretical)** – Proposed Modified Newtonian Dynamics (MOND) as an alternative to dark matter. MOND introduces a new constant acceleration scale $a_0 \sim 1.2 \times 10^{-10} \text{ m/s}^2$. At accelerations $a \gg a_0$, gravity effectively scales as $1/r$ rather than $1/r^2$, so a test particle feels $a \sim \sqrt{GM a_0}/r$, giving asymptotically flat rotation curves without any extra mass. As one review puts it, “flat rotation curves are automatically explained” by MOND’s $1/r$ regime ⁸. In practice MOND fits rotation curves with essentially one free parameter (the stellar mass-to-light ratio), and it naturally predicts the observed RAR critical acceleration. While MOND is not a mainstream explanation, it remains a well-known theoretical alternative.
- **Scarpa (2012, *Int. J. Mod. Phys. D*; review)** – Introductory review of MOND and its application to galaxy dynamics. Scarpa (building on Milgrom’s idea) reiterates that MOND’s interpolating function is chosen so that $a \gg a_0$ gives $a \propto 1/r$, automatically yielding flat outer curves ⁸. He contrasts this with dark-matter fits that have many free parameters. This review summarizes many fits of MOND to actual rotation data, showing that in many cases MOND matches the curves well, though challenges remain (especially in clusters).
- **Mannheim (2006, *Prog. Part. Nucl. Phys.*; review/theoretical)** – A theoretical review of *alternative gravity theories* (including Conformal Gravity) as solutions to the dark-matter problem. Mannheim argues that the assumption of unmodified gravity on all scales may be unwarranted. He identifies **three generic features** of such alternatives: (1) a universal acceleration scale below which Newtonian gravity fails, (2) a global cosmological term that affects local galactic motions (mimicking a dark halo), and (3) a different coupling of the cosmological constant. In effect, these theories can reproduce flat rotation curves (and other galaxy scaling laws) by invoking a new fundamental scale or background effect, rather than particle dark matter ⁹. Mannheim’s review emphasizes how flat rotation-curve phenomenology (including the RAR) might arise in these frameworks.

Each of the above studies is peer-reviewed and widely cited. Together they establish that **spiral galaxy rotation curves remain high at large radii**, requiring far more mass than is seen in stars and gas. The observational papers consistently find that virtually no galaxy shows the Keplerian decline expected from the light distribution alone ¹ ². Review articles (Sofue & Rubin 2001) and surveys (Persic *et al.* 1996)

summarize this evidence, while theoretical explorations (Milgrom 1983; Mannheim 2006) show how modified dynamics can mimic it ⁸ ⁹. In all cases, the key finding is that the flatness and detailed shapes of rotation curves imply a **mass discrepancy** that is most straightforwardly explained by extended dark halos around galaxies.

Sources: Classic and modern studies of galaxy rotation (e.g. Freeman 1970; Rubin & Ford 1970s; Bosma 1981; Begeman *et al.* 1991; Persic *et al.* 1996; Catinella *et al.* 2006; de Blok *et al.* 2008; Lelli *et al.* 2017; Noordermeer *et al.* 2007) and review/theory articles (Sofue & Rubin 2001; Scarpa 2012; Mannheim 2006) as cited above.

¹ Galaxy Disks

https://www.astro.rug.nl/~vdkruit/jea3/homepage/AnnRev_vdKruit-Freeman.pdf

² ⁴ oup.silverchair-cdn.com

<https://oup.silverchair-cdn.com/article-minimal/1014139>

³ [astro-ph/9506004] The Universal Rotation Curve of Spiral Galaxies: I. the Dark Matter Connection

<https://arxiv.org/abs/astro-ph/9506004>

⁵ [0810.2100] High-Resolution Rotation Curves and Galaxy Mass Models from THINGS

<https://arxiv.org/abs/0810.2100>

⁶ [1610.08981] One Law To Rule Them All: The Radial Acceleration Relation of Galaxies

<https://arxiv.org/abs/1610.08981>

⁷ Rotation Curves of Spiral Galaxies | Annual Reviews

<https://www.annualreviews.org/content/journals/10.1146/annurev.astro.39.1.137>

⁸ arxiv.org

<https://arxiv.org/pdf/astro-ph/0601478>

⁹ [astro-ph/0505266] Alternatives to Dark Matter and Dark Energy

<https://arxiv.org/abs/astro-ph/0505266>